SAUDI DIGITAL ACADEMY الأكاديمية السعودية الرقمية

CODING DOJO ACADEMY

Big Data & AI Bootcamp

Pipelines Project

Flight Status Prediction using Machine Learning

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I. Introduction to the Dataset

How many of us have experienced flight delays or cancellations? Were you always questioning, "What caused them to be late?" What variables might cause so many flight delays and cancellations? Which airlines have a history of the most delays, so we can avoid them in the future? After these catastrophes, were airline stocks impacted by the delays? In this report, this is what we are attempting to explore.

Dataset Title: Flight Status Prediction

Dataset Author: Rob Mulla

Resource: Kaggle 1].

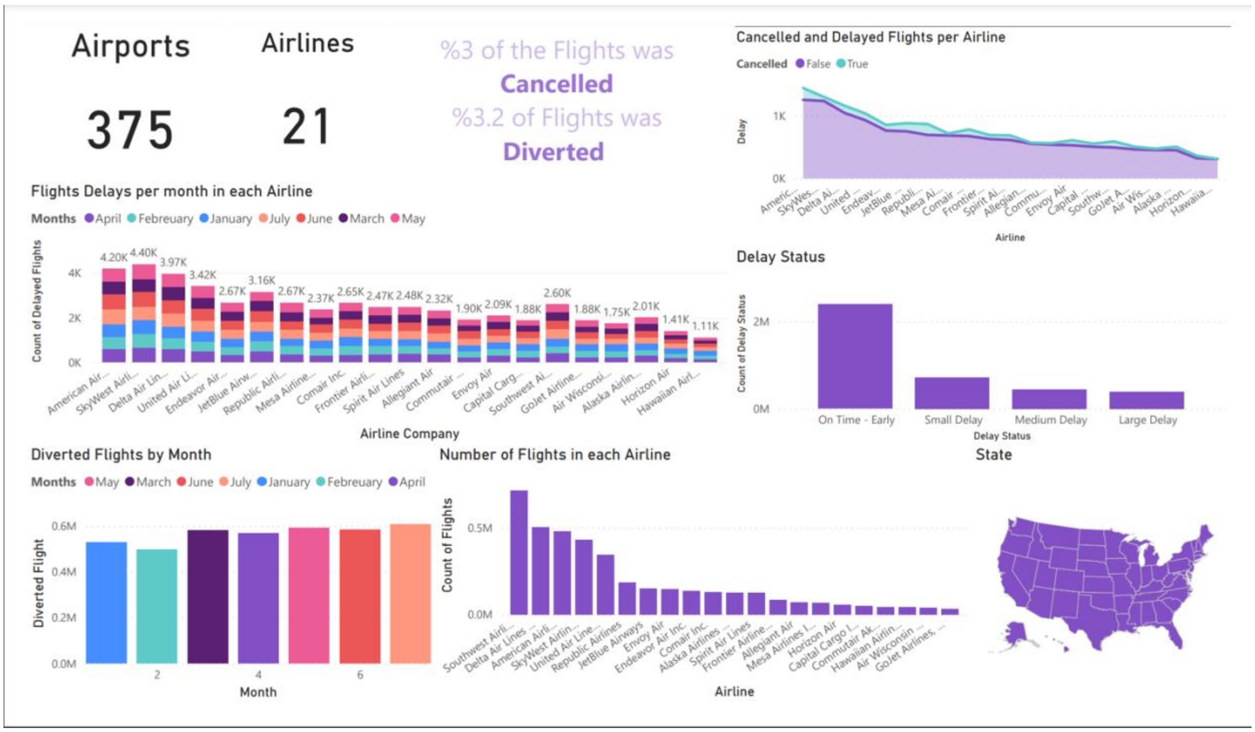
This dataset provides all flight information, including cancellations and delays, per airline, with +40M Flights and 61 Columns, dating back to January 2022. The EDA consumed most of our work due to the enormous amount and dimension of the dataset. The features consisted primarily of ID's, information regarding delay times, and diverted flights, the ones we were inter. Consequentially, the subsequent questions demonstrate our interest in the dataset:

- Is the flight going to be delayed?
- How many airports and airlines?
- How many flights were canceled and delayed per airline?
- Does the month have an effect on the delayed and diverted flights?
- What is the distribution of the different delay statuses?
- How many flights are there on each airline?

Table 1 Dataset Description

CRSDepTime	CRS Departure Time.
WheelsOff	Wheels Off Time.
WheelsOn	Wheels On Time.
Distance	Distance between airports (in miles).
DistanceGroup	Distance Intervals, every 250 Miles, for Flight Segment.
TaxiOut	Taxi Out Time.
TaxiIn	Taxi In Time.
ArrDelayMinutes	Difference in minutes between scheduled and actual arrival time. Early arrivals → 0.
DepDelayMinutes	Difference in minutes between scheduled and actual departure time. Early departures → negative numbers.
DepDelay	Difference in minutes between scheduled and actual departure time. Early departures → 0.
DepDel15	Departure Delay Indicator, 15 Minutes or More.
DepartureDelayGroups	Departure Delay intervals.

II. Exploratory Data Analysis



III. Feature Engineering

Before proceeding with feature engineering, it was necessary to alter a few phases. Due to the vast number of columns, we selected the ones that aided in our analysis and pair-plotted them to determine their correlation.

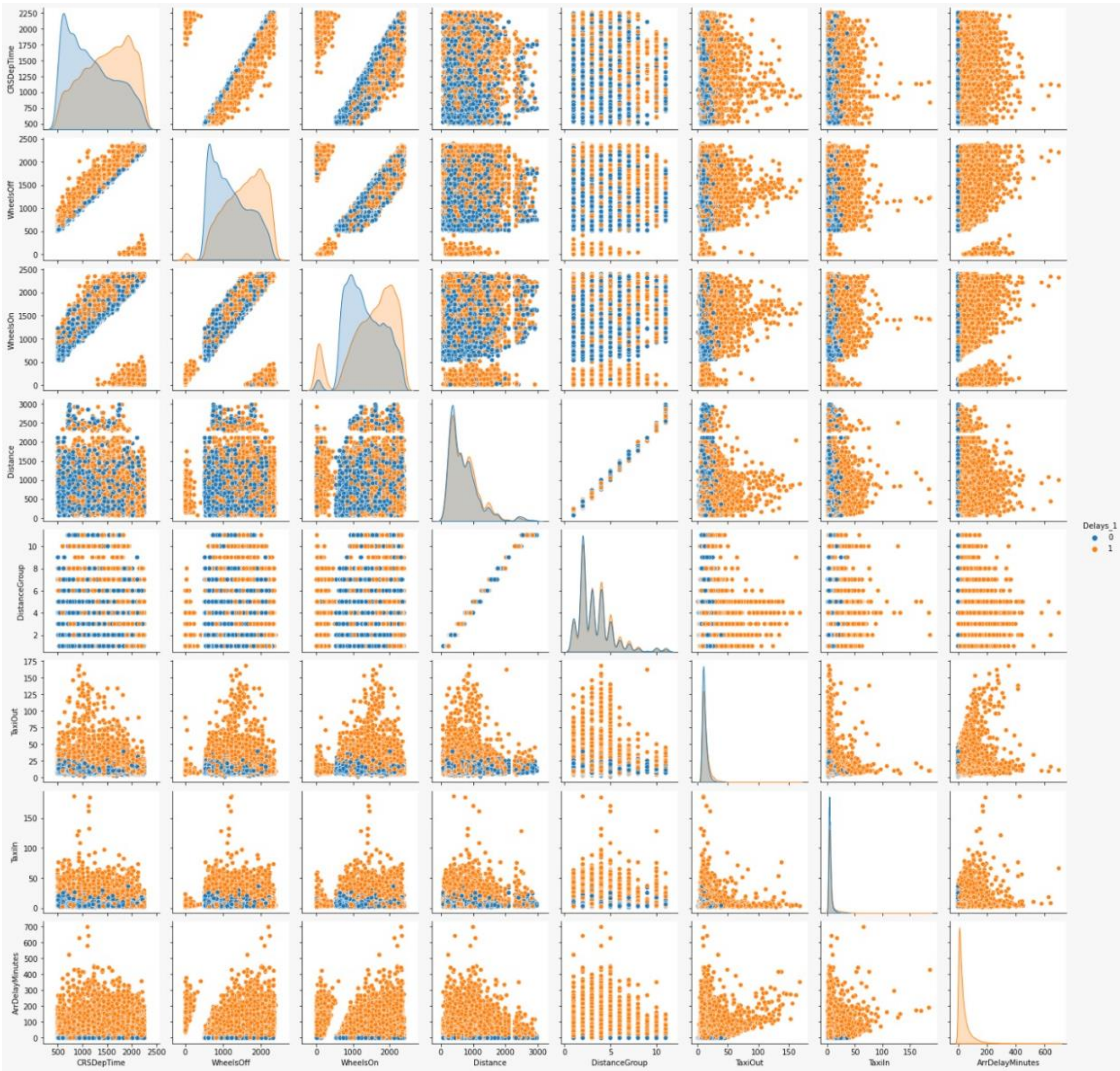


Figure 1 Flight Status Pair Plot

Then, during the phases of feature engineering, we traversed various stages. First, we removed the unnecessary columns from the dataset. Second, we have dealt with the missing data for each model and each feature set. Occasionally, we had to exclude the null values since they reflected canceled flights, which contradicted our target, the delayed flight target. Sometimes, we used the mean of the column values to fill in the missing values. Then, for the classification model, the continuous variable 'ArrDelayMinutes' was binned into two possible categories:

```
Delays_1 = []
for row in df['ArrDelayMinutes']:
        if row == 0: Delays_1.append('NoDelay')
        elif row > 0: Delays_1.append('Delay')
```

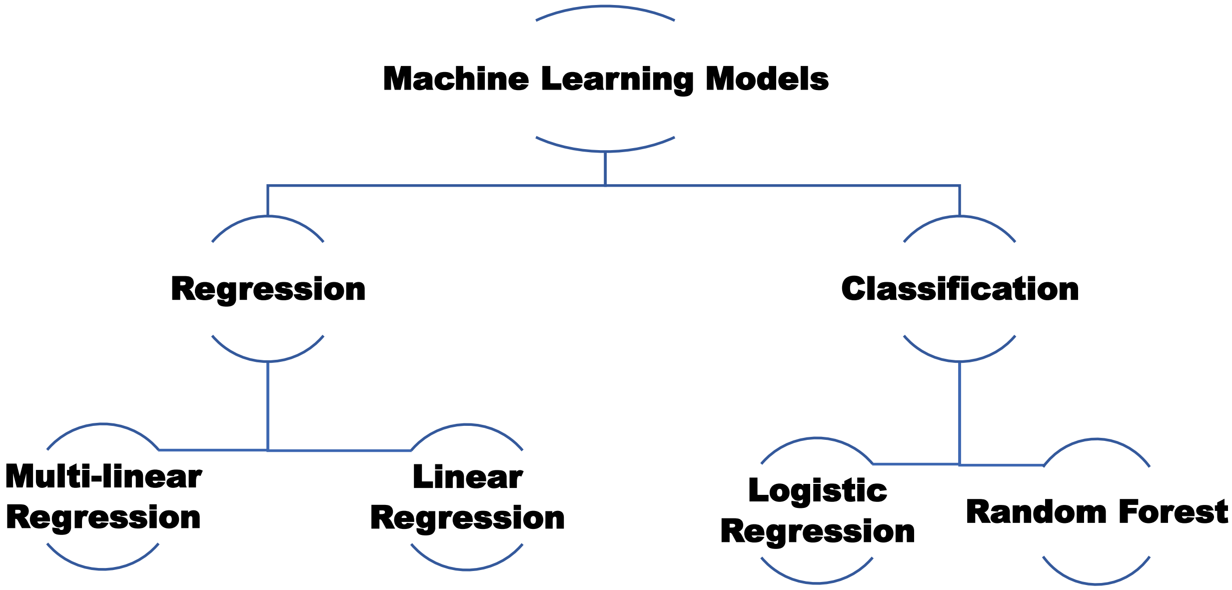
After the binning, we performed ordinal encoding to the same column.

```
#ordinal encoder
df['Delays_1'] = Delays_1
df['Delays_1'].replace(['NoDelay', "Delay"], [0, 1], inplace=True)
```

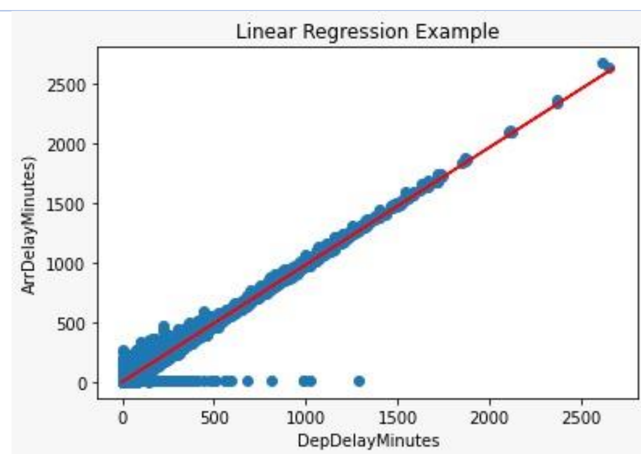
Finally, after all these steps, we scaled the numerical features using standard scaling, in order to enhance the model performance and split the data for the prediction step.

IV. Predictions

In our prediction, we wanted to see the probability of a flight delay. Thus, we have created multiple models to predict this using different model types and different feature sets.



<table>
<tr><td rowspan="4">Classification</td><td rowspan="4">Logistic Regression</td><td>Features</td><td colspan="2">CRS Departure Time, Wheels Off, Wheels On, Distance, Distance Group, Taxi Out, Taxi In</td></tr>
<tr><td>Target</td><td colspan="2">Delay</td></tr>
<tr><td rowspan="2">Confusion Matrix</td><td>119184</td><td>0</td></tr>
<tr><td>3090</td><td>68987</td></tr>
</table>

<table>
<tr><td rowspan="4"></td><td rowspan="4">Random Forest</td><td>Features</td><td colspan="2">CRS Departure Time, Wheels Off, Wheels On, Distance, Distance Group, Taxi Out, Taxi In</td></tr>
<tr><td>Target</td><td colspan="2">Delay</td></tr>
<tr><td rowspan="2">Confusion Matrix</td><td>119184</td><td>0</td></tr>
<tr><td>0</td><td>72077</td></tr>
<tr><td rowspan="5">Regression</td><td rowspan="4">Linear Regression</td><td>Features</td><td colspan="2">Departure Delay Minutes</td></tr>
<tr><td>Target</td><td colspan="2">Arrival Delay Minutes</td></tr>
<tr><td>Mean Absolute Error</td><td colspan="2">5</td></tr>
<tr><td>Result</td><td colspan="2">
</td></tr>
<tr><td>Multi-linear Regression</td><td>Features</td><td colspan="2">Departure Delay Groups, Departure Delay by 15 Minutes, Departure Delay, Departure Delay Minutes</td></tr>
</table>

		Target	Arrival Delay Minutes
		Mean Absolute Error	5
		R2 Score	96%

V. Conclusion

On the basis of the model findings, we can conclude that our models are capable of predicting with 100 percent accuracy, using Random Forest, whether there would be a delay based on the mentioned features. If the departure delay minutes are known, the model can also estimate the arrival delay minutes with a mean absolute error of five. Lastly, the model can accurately forecast arrival delay minutes using the aforementioned various features. In addition, we have determined that the dataset is unsuitable for machine learning and prediction because the majority of its features result in data leakage in the model. However, it may only be suited for dashboard visualizations.

Resources

1. [Dataset link]
2. **Dashboard**
3. [Presentation link]